While shipping is a carbon efficient transport mode, given that roughly 90% of the world trade is carried by ships, the negative impact of shipping on human health and the natural environment is significant. One of the attempts being made by the shipping industry to reduce its environmental impact is to use liquefied natural gas (LNG) as a marine fuel. Various incentives and motivations, including laws and regulations, to reduce shipping's exhaust emissions are being adopted. One of the emission reduction attempts is the use of LNG as a marine fuel. LNG, as a marine fuel, produces virtually 0% sulfur dioxide emissions. Moreover, compared with conventional marine bunker fuels, LNG has the potential to reduce carbon dioxide emissions by 10% to 20%, nitrogen oxide emissions by up to 80% or 90%, and particulate matter emission by 98% to 100%.

For some types of ships, this alternative fuel has become a strong competitor to traditional bunker fuels and, for those ship types that were not traditionally associated with LNG, the utilization of LNG as a marine fuel has started to emerge.

The MARPOL Convention addressed five types of pollutants through its five original Annexes: Oil, Noxious Liquid Substances, Packaged Harmful Substances, Sewage, and Garbage. In 1997 through a protocol that entered into force in 2005, Annex VI was added to deal with the problem of air pollution caused by exhaust emissions from ships.11 Annex VI establishes limits on sulfur oxide and nitrogen oxide emissions from ship exhausts, prohibits deliberate emissions of ozone-depleting substances, and provides for emission control areas in which more stringent standards apply

In 2008 amendments to Annex VI were made, and they became legally effective from 2012.13 One of the amendments reduces the global sulfur cap to 3.50% with a potential further decrease to 0.50% in January 2020, subject to a feasibility review to be completed no later than 2018.14 Stricter limits were also introduced for sulfur oxide emission control areas (SECAs),15 where a limit of 1.00% became applicable from 1 July 2010 and a 0.10% limit to apply from 1 January 2015. The 4.5% cap on sulfur content was considered groundbreaking when adopted,16 and the move to 3.50% as a general limit is significant. The potential move to a 0.50% limit in 2020 constitutes a major incentive for the widescale adoption of LNG as a marine fuel.

LNG CONTRIBUTION TO ANNEX VI